

Response to Request for Information (RFI) on AI Action Plan

Kaushik Chowdhury, University of Texas at Austin

Yingbin Liang, The Ohio State University

Jia Liu, The Ohio State University

Sanjay Shakkottai, University of Texas at Austin

Ness Shroff, The Ohio State University

All authors are affiliated with the NSF AI-EDGE Institute

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Introduction

The United States has long been the global leader in artificial intelligence and communication networking – two of the most foundational fields of the IT industry. This leadership has driven innovations that have shaped the modern digital world and generated industries that have already created trillions of dollars in economic growth and provided employment for millions of US workers. These technologies are going to continue to improve people's lives and be critical for future economic growth, job creation, national security, and national defense. However, the rapid surge in AI and 5G investments worldwide, particularly in countries heavily investing in next-generation connectivity and AI infrastructure, poses a significant challenge to U.S. dominance in these critical domains. Recent breakthroughs in generative AI, exemplified by large language models (LLMs) like GPT and DeepSeek, have demonstrated unprecedented capabilities in reasoning, adaptation, and real-time interaction. These AI advancements are reshaping industries, including networking, and redefining human-machine collaboration. Therefore, integrating AI and networking technologies such as 6G/7G in a closely interdependent manner is imperative to ensure continued U.S. leadership in these fields.

To this end, two critical and complementary research challenges must be addressed: *AI for Networks* and *AI on Networks*. The former would focus on leveraging AI to develop next-generation intelligent networks that are more adaptive, efficient, and capable of self-optimization – enabling real-time traffic management, enhanced cybersecurity, and AI-assisted 6G/7G innovations. The latter needs to concentrate on designing and facilitating distributed AI systems that can operate seamlessly on decentralized edge networks leveraging advanced networking technologies to optimize AI model efficiency, reduce latency, and enhance robustness, safety, and privacy of AI computation at the edge. These heterogeneous edge devices that will form the network-edge are being created in the billions and are expected to become a major driver of innovations in health, medicine, drug discovery, traffic engineering, and automation. By fostering a synergistic relationship

between AI and networking, the U.S. can create a virtuous cycle of innovation, where breakthroughs in AI models drive networking advancements, and improved network infrastructures, in turn, enable more scalable and efficient AI deployments. This integrated vision is essential for securing U.S. leadership in both AI and networking, ensuring US technological dominance in an era increasingly defined by intelligent and interconnected systems.

AI for Networks: Transforming Network Intelligence

As the growth of wireless communication accelerates, future networks will increasingly rely on edge computing, where computational and data-intensive tasks occur closer to the user. These networks comprise complex distributed systems with diverse network elements, including heterogeneous software and hardware components. Traditional network management approaches, relying on heuristic designs and domain-specific knowledge, are increasingly inadequate in managing the scale, dynamism, and uncertainty of modern networks.

The astonishing successes of AI, particularly in Machine Learning (ML) and Generative AI, provide a transformative opportunity to design next-generation networks that are not only more efficient but also intelligent and self-optimizing. Recent advancements in foundation models, such as Large Language Models (LLMs) like GPT and DeepSeek, as well as generative AI techniques including diffusion models, have demonstrated remarkable capabilities in reasoning, decision-making, and long-term sequence modeling. These capabilities can be harnessed to develop AI-driven edge networks that can dynamically learn, predict, and optimize network behaviors with minimal human intervention.

By leveraging AI's ability to model long-term dependencies and adapt in real time, we can transition from rigid, manually managed networks to autonomously operating intelligent systems. The incorporation of AI agents into networking infrastructures enables automated optimization of routing, traffic engineering, network security, and resource allocation—all of which are crucial for handling the increasing complexity of edge computing environments. These AI agents, powered by LLMs and multi-modal AI models, can understand, infer, and optimize network conditions dynamically, ensuring efficient and robust performance under highly variable and uncertain conditions.

Moreover, as edge networks predominantly consist of wireless devices, their distributed nature demands a decentralized AI-driven management paradigm. A distributed AI control plane would fundamentally transform both current and future network architectures in two key ways:

1. **Optimizing Existing Networks** – Current networking architectures can be revitalized by replacing antiquated management frameworks with self-learning AI agents that continuously interact, analyze, and optimize network control and data planes. These agents can intelligently balance loads, predict failures, and dynamically adjust resources without human intervention.

2. Designing Future Networks – The next generation of networks will be built with an AI-driven, distributed intelligence at their core, ensuring security, efficiency, and adaptability. Through co-designing the data, control, and management planes, AI will enable fully autonomous, self-healing, and resilient networking ecosystems that can proactively respond to threats, congestion, and operational challenges in real time.

By integrating AI at every layer of network design, from intelligent traffic prediction and anomaly detection to generative AI-assisted self-configuration and automated network planning, the future of networking will be defined by adaptive, self-optimizing, and ultra-efficient infrastructures. This vision will not only enhance performance and security but also ensure that AI-powered networks can scale seamlessly to meet the demands of an increasingly connected world.

AI on Networks: Revolutionizing Distributed AI

Distributed AI will play a critical role in the future of AI, particularly as AI-driven technologies has become integral to everyday life and smart mobile devices have proliferated in the last decade. As a result, we are witnessing a significant portion of AI-powered applications—ranging from personal assistants to intelligent IoT devices, real-time analytics, and autonomous systems—are increasingly shifting to the edge and personal devices rather than relying solely on centralized cloud infrastructures. The recent breakthroughs in large-scale generative AI models, such as GPT, DeepSeek, and diffusion models, are predominantly trained and controlled by large corporations that can afford the extensive computational and data resources required for their development.

To democratize research and development of such transformative generative AI models and enable flexible, secure, and personalized AI experiences for individuals, a shift toward *networked yet distributed* AI training and inference is not only highly desirable but also necessary. We envision that future AI applications based on generative AI models will need to be deployed and fine-tuned across decentralized, heterogeneous, and privacy-sensitive environments rather than being confined to a few centralized data centers. Enabling such distributed AI mechanisms not only makes AI more scalable, accessible, and trustworthy but also ensures that powerful AI models are available to all—not just large corporations with extensive computational infrastructure.

As AI shifts toward edge-first deployment, a fundamentally different landscape arises. Future edge networks will consist of resource-constrained devices that introduce significant challenges: (i) limited memory and computational power at edge nodes, requiring efficient model compression, quantization, and adaptive training techniques; (ii) straggler problems, where heterogeneous devices experience varying delays, making traditional synchronous training inefficient; (iii) slow and unreliable communication links, necessitating network-aware AI algorithms that minimize communication overhead; and (iv) imbalanced, heterogeneous, and privacy-sensitive datasets, requiring novel approaches such as federated learning, differential privacy, and self-supervised adaptation. Given these constraints, it is essential to develop scalable, network-aware

distributed AI algorithms that jointly optimize computation, communication, and data utilization in edge settings. These algorithms must be adaptive, fault-tolerant, and capable of continuous learning in dynamic environments.

Since distributed AI algorithms inherently operate over networked infrastructures, it is natural for them to leverage and co-design with the underlying network to achieve optimal performance. This motivates a fundamental shift: instead of networks being designed separately from AI, future networks must be re-engineered to become AI-aware, optimizing resources dynamically to support intelligent applications. To this end, future AI-aware networks must be designed to intelligently allocate bandwidth and processing power to optimize real-time AI inference and training; dynamically optimize data movement and storage to ensure the right information reaches the right AI models at the right time; incorporate AI-driven network orchestration to predict congestion, preemptively reconfigure resources, and enhance resilience to failures; and enable privacy-preserving AI architectures that support secure federated learning and encrypted model training across decentralized nodes. By bridging the gap between network optimization and distributed AI, we can ensure that the next generation of AI-powered systems operates with maximum efficiency, scalability, and adaptability—bringing powerful AI capabilities directly to individuals and decentralized communities rather than being monopolized by centralized entities.

Explainability of AI: Ensuring Reliable AI for and on Networks

The explainability of AI/ML models (XAI) is crucial for their adoption in wireless networks, yet research in this area remains limited. Hence, there is need to pursue foundational knowledge that leads to interpretability of such models for network performance analysis, resource management, 5G service classification in the upper layers of the stack, as well as spectrum analysis and sensing, at the physical layer. As publicly available RF datasets grow, tools that offer opportunities for rapid data visualization and on-site human-level understanding of the features that the trained ML models are learning will become increasingly important. Other domains, such as image processing, have pioneered methods like Class Activation Mapping (CAM) that highlights regions of images that the model is focusing on during training, which could be adapted for RF data, e.g., image-like spectrograms, for improving trust in automated wireless systems. When the number of input features to a ML model is very large, there are many challenges of storing, processing and relaying these features to/from the wireless edge. In such cases, approaches like SHapley Additive exPlanation (SHAP) can be utilized to identify the most discriminative features as determined by the model, which can then be used to reduce the overhead of collecting an exhaustive set of features at the edge.

There is a need for certifying ML models to mitigate risks posed to network functionalities in case of failures once they are deployed. This is challenging since (i) testing on every exhaustive case is infeasible and (ii) the data used for model training may not be available. While advancing the foundational research for robust ML models that are highly resilient to both naturally evolving environmental conditions and security breaches, it is also of high interest in developing ML-guided test and measurement practices. These efforts will

lead to ML-generated tests for previously trained models and provide benchmarks to measure the performance against alternate deterministic or heuristic solutions. One promising approach is to use gradient-based optimization, since it avoids spanning the exhaustive set of all wireless environment and channel conditions; instead, it generates the next test in real time to further probe specific configurations that offer the highest risk of failure. Such radically different methodologies, co-developed with inputs from leading US-based test and measurement companies will lead the way in developing a new science for verifying and validating model performance.